

COURSE TITLE : RUBBER PRODUCTS
COURSE CODE : 4094
COURSE CATEGORY : A
PERIODS/WEEK : 6
PERIODS/SEMESTER: 84/4
CREDITS : 5

TIME SCHEDULE

Module	Topics	Period
1	Moulded and expanded rubber products	24
2	Extruded , calendared and miscellaneous rubber products	20
3	Rubber to metal bonded rubber products and rubber coated textile products	20
4	Assembled rubber products	20
TOTAL		84

COURSE OUTCOME

Sl.	G.O	Student will be able to
1.	1	To know the moulded and expanded rubber products
	2	To know extruded, calendared and miscellaneous rubber products
2.	1	To understand the rubber to metal bonded rubber products and rubber coated textile products
3.	1	To know about assembled rubber products

SPECIFIC OUTCOME

MODULE I MOULDED AND EXPANDED RUBBER PRODUCTS.

1.1.0 To know the moulded and expanded rubber products .

- 1.1.1. Classify rubber products according to their applications
- 1.1.2. Explain the different steps involved in the production of compression moulded products
- 1.1.3. Design of compounds for Auto mobile bushes mats ,paper weights, I.B caps , nipples and Hotwater bags .
- 1.1.4. Define hard rubber (Ebonite)
- 1.1.5. Explain the compounding and production of hard rubber.
- 1.1.6. Explain various finishing operations for moulded rubber products.
- 1.1.7. Define cellular rubbers and classify them.

- 1.1.8. Compare merits and demerits of different blowing agents
- 1.1.9. Describe the manufacture of MC ,Hawai sole and sponge with typical recipe.
- 1.1.10. Identify the major defects and remedies of compression moulded products

MODULE II EXTRUDED ,CALANDERED AND MISCELLANEOUS RUBBER PRODUCTS

2.2.0 To know extruded and calandered rubber products

- 2.2.1. List various extruded and calandered rubber products.
- 2.2.2. Explain the manufacture of automobile channels and rubber tubings with typical recipe.
- 2.2.3. Distinguish between supported and unsupported sheets.
- 2.2.4. Explain the manufacture of hospital sheeting and supported sheeting with typical formulations .
- 2.2.5. Describe profile calendering with examples .
- 2.2.6. Identify the major defects and remedies of extruder and calandered products .
- 2.2.7. Explain the manufacture of play ball, tennis ball and golf ball with special reference to formulations and principle of expansion.
- 2.2.8. Explain the manufacture of gaskets and seals with reference to formulations .
- 2.2.9. Describe the manufacture of different types of rubber foot wears .
- 2.2.10. State the advantages and disadvantages of rubber footwear over leather foot wear.

MODULE III RUBBER TO METAL BONDED RUBBER PRODUCTS AND RUBBER COATED TEXTILE PRODUCTS

3.3.0 To understand the rubber to metal bonded rubber products and rubber coated textile products

- 3.3.1. List rubber to metal bonded products .
- 3.3.2. Explain the principle and steps involved in the production of rubber to metal bonded products .
- 3.3.4. Give typical recipes for type writer roller, paper rolls ,textile roller and rice polisher .
- 3.3.5. List various rubber coated products.
- 3.3.6. Explain the different steps involved in the spreading process.
- 3.3.7. Explain the adhesive treatment for nylon 6, nylon6 and polyester.

MODULE IV ASSEMBLED RUBBER PRODUCTS

4.4.0 To know about assembled rubber products

- 4.4.1. Classify beltings according to their functions.
- 4.4.2. Explain the components, compounding and manufacture of conveyor and transmission belts .
- 4.4.3. Describe building and curing of flat transmission and V- belts.
- 4.4.4. Classify hoses according to their construction and end users .
- 4.4.5. Explain the components, compounding and manufacture of braided hose, knitted hose, woven hose ,spiral reinforced and circular woven hose .
- 4.4.6. Explain the production of molded and mandrel cured hoses.
- 4.4.7. Compare wrapped and braided hose .
- 4.4.8. Classify cables according to their end use.
- 4.4.9. Describe the components ,manufacturing and curing of different types of cables .

4.4.10. Design compounds for the various components of cables.

CONTENT DETAILS

MODULE I

Molded and Expanded rubber products

Classification of rubber products – steps involved in compression moulded products - Production methods, compound design - Automobile bushes-Door mats, Tea-Mats, Paperweights, IB Caps, Door bushes, Nipples Hot water bags-Chemistry of hard rubber products (Ebonite). Method of production and compound design for battery boxes -Finishing of rubber mouldings. **Expanded Products:** Types of blowing agents and their comparison- Compounding and Molding of Cellular rubber products, MC, Hawaii sponge -Typical formulations. Defects and remedies of molded products.

MODULE II

Extruded, Calendered and Miscellaneous rubber Products: Extruded rubber products -Automobile treads,- channels-, inner tubes, -LPG tubes etc- Compound design, Typical formulations, Methods of manufacture.

Calendered rubber Products

Supported and unsupported sheeting- Hospital sheeting, -production method -Profile calendered products,. Production of supported sheets. Design criteria-compound design typical formulations. Defects and remedies of the above products **Sports Goods:** Play ball, Tennis balls, Golf Balls, - Components, compound design, method of manufacture, typical formulations. **Miscellaneous Products:** Oil seal, Gaskets- Compound design, Manufacture and Typical formulations **Footwear:** Components of footwear, materials, methods of manufacture. Special purpose footwear and their applications.

MODULE III

Rubber to Metal bonded products and coated textile products -

List major rubber to metal bonded products -Principle of bonding rubber to metal and textile fibers. Production of rubber covered rollers –type writer roller –paper roller – textile roller –rice polisher . metal preparation- compound preparation, assembling curing, finishing-Tank lining- Production method, compounding, compound design.

Rubber coated textile products - Rexene, Tarpaulin and other rubber coated textile, rubber lining, spreading. Design principles of spreading, compounding and typical formulations. Adhesive treatment for Nylon 6- Nylon 66- Polyester-Adhesive composition and their preparations.

MODULE IV

Assembled rubber products.

Beltings: Classification -Conveyor belts- Transmission and V- belts: Components manufacturing techniques, Batch and continuous curing, Compounding, Typical formulations.

Hoses: Classification according to manufacture and end use. Components. Molded hose, Mandrel cured hose- comparison. Braided hose, Knit hose, Woven hose, Spiral reinforced hose, Circular Woven hose, Handmade hose, Radiator hose, Garden hose, Petrol hose, Sand blasting hose, Typical formulations, method of manufacture.

Cables: Classification, Power cables and their components, manufacturing methods, Curing- Compound design- Typical formulations.

REFERENCE BOOKS

1. Rubber Technology And Manufacture- C.M.Blow (Publisher – Butterworth Hainemann Ltd,
1982 Edition)
2. Rubber Technology -Maurice Morton (Publisher – Springer, 1999 Edition)
3. Vanderbilt Rubber Hand book -R.T Vanderbilt (Publisher – R.T Vanderbilt, 1990 Edition)